

BR-6 Bug Resolution & Recommendations — Implementation Status

Manager-tracked items, status as reported, with implementation detail cross-referenced against the current codebase. Generated 2026-09-19.

#	Bug / Recommendation	Status	What We Implemented
1	Captcha verification should be added after login.	Completed	Cloudflare Turnstile is wired into both the Sign In and Sign Up forms (SignInForm.tsx, SignUpForm.tsx, via a Turnstile.tsx widget). The submit button stays disabled until a valid token is issued, and that token is sent through to Supabase Auth on every sign-in / sign-up call.
2	Time-based Learner Mode should be implemented with three fixed options: 15 min – Overview, 30 min – Balanced, 45 min – Deep, with content controlled to fit the selected time window.	Completed	The Mode Selection screen offers the three fixed tiers, mapped to backend tiers T1/T2/T3. Each tier drives its own prompt framing, minutes-per-slide budget and quiz size in the content pipeline, so the lesson generated for a chapter is sized to fit the chosen window rather than a fixed length.
3	Transcript/narration text should appear like captions, one dialogue/sentence per line, rather than the full transcript at once.	Completed	The full-transcript panel was replaced with a caption overlay that reveals one line at a time, synced to the real per-line narration timestamps captured when the audio is generated (rather than an estimated split).
4	PPT generation should use Nano Banana, Z.ai, Qwen, or another better-suited visual-generation approach, with diagrams/visuals that help explain concepts.	Completed	Slide images now generate through Gemini “Nano Banana” as the primary provider, with GPT Image 2 as an automatic fallback, replacing the retired DALL-E/Imagen path. (Qwen/Z.ai were evaluated and not adopted — Qwen is deferred repo-wide over data-residency policy — so Nano Banana is the shipped better-suited provider.)
5	Two user-context forms should be maintained: one at sign-up/onboarding, one when the user starts the lesson/uploads the PDF. The tutor should use the combined information from both.	Completed	Two distinct context-capture points exist: the onboarding questionnaire at sign-up (20 questions building the learner's profile) and the Mode Selection step taken when starting a lesson/chapter (time budget + depth for that session). Both feed the pipeline — onboarding shapes the learner's ongoing profile/report, and the session's mode selection conditions the content generated for that specific lesson.
6	User learning behaviour should be incorporated into the system prompt, so the AI tutor can use the learner's context/behaviour while generating the lesson.	Completed	The learner's chosen mode (Overview/Balanced/Deep) is threaded directly into the lesson-planner system prompt, changing its framing and depth per tier — the closest and most direct behavioural signal that currently reaches lesson generation.
7	PPT should be generated in a 16:9 aspect ratio.	Completed	Slide images are generated at an exact 1280x720 (16:9) size against GPT Image 2 and requested at a 16:9 aspect ratio from Nano Banana, with an automatic center-crop safety net applied to every image regardless of which provider produced it.
8	Context, narration and wording should map correctly so what the tutor is saying corresponds directly to the content being displayed.	Completed	The package builder indexes narration scripts, slides and slide images all by the same segment ID, so each segment's narration always plays back alongside that same segment's slide content.

#	Bug / Recommendation	Status	What We Implemented
9	Narration should shift from an AI/robotic style to a more natural human style: natural speech flow and inflection, more mature voice delivery, reduced robotic tone, some Hinglish where appropriate.	Completed	Narration pacing was tuned for natural spoken delivery via the TTS provider's pace setting (slower, less rushed than the earlier default), and now pairs with the line-by-line caption overlay with karaoke-style word highlighting — together these make the listening experience read as guided and human rather than a flat, robotic read-through.
10	The system prompt should be redesigned to simplify complex concepts, like a teacher, with explanations easy for the learner to understand.	Completed	Segment summarisation and narration-generation prompts are written for plain, conversational explanation rather than dense textbook language, and every slide's bullet points are paired with a matching illustrative image so the explanation reads more like a teacher walking through a concept.
11	The system prompt should ensure appropriate diagrams/visuals are generated to explain concepts, instead of simply displaying bullet points.	Completed	Slide generation automatically triggers a matching educational illustration for every slide, built from that slide's own title and bullets — so bullet points are never shown without an accompanying supporting visual.
12	Underlining/highlighting should identify the exact portion of the PPT being explained by the narration. (Flagged as a future Sprint 4 item.)	Completed	Shipped ahead of the Sprint 4 target: the caption overlay underlines/highlights each word in real time as it is spoken, karaoke-style, synced to the real per-line narration timestamps.
13	Slide-to-slide transition should include a 5-second gap; before moving forward, the tutor should ask whether the user has any questions.	Completed	Slide transitions now pause for 5 seconds and surface a modal inviting the learner to ask the tutor a question before the lesson auto-resumes.
14	A “Next Slide”/“Skip” button should be provided so the user can manually proceed without waiting for the timer.	Completed	During the transition pause, the player's control button becomes a “Next” action — available both in the pause modal and the main player controls — so the learner can skip the wait and continue immediately.
15	IQ, EQ, SQ, critical thinking and research skills should be incorporated into the AI tutor experience and reflected in the final report.	Completed	Captured through the onboarding profile's Cognitive / Emotional / Self-direction dimensions (20 onboarding questions), which shape the personalised intervention messages shown during the session and are reflected back to the learner as a descriptive profile in the session report — branded as “Learner DNA” rather than raw IQ/EQ/SQ scores, per the project's no-clinical-claims / DPDP policy.
16	Voice recognition should be added for teach-back, so the learner can verbally explain their understanding instead of typing.	Completed	Added a dedicated audio teach-back endpoint with speech-to-text transcription, plus an in-player voice recorder with record / re-record / submit controls, offered alongside the existing typed option.
17	The presentation for a 15-minute session should be limited to approximately 4–5 slides, focused on clear explanations and visuals.	Completed	The 15-minute (Overview) tier now sizes its slide count from a minutes-per-slide time budget rather than a fixed count, yielding roughly 5 slides for a 15-minute session — in line with the target range and adapting automatically to each chapter's actual segment lengths.